

Cyber Threat Intelligence Report



MISP

MISP (Malware Information Sharing Platform) is an open-source tool used for collecting, storing, and sharing threat intelligence. It helps security teams and organizations keep track of:

- Malware information
- Cyber attack details
- Financial fraud data
- Other security-related intelligence

Information Sharing in MISP

MISP allows sharing in different types of communities:

- Closed → within one organization or small group.
- Semi-private → between selected trusted partners.
- Open/Public → shared widely with many organizations.

The shared threat data can be used by other security tools, such as:

- NIDS (Network Intrusion Detection Systems)
- Log analysis tools
- SIEM platforms (for monitoring and detection)

Usage:

- Central place to collect and organize threat data.
- Helps in correlating attacks (e.g., same IP used in two phishing campaigns).
- Reduces false positives with features like warninglists.
- Allows sharing with partners using standards (STIX, TAXII).
- Supports collaboration, so different teams can learn from each other's incidents.

MISP Workflow

1. Data Ingestion (Input):

Threat intel comes from manual analyst input, feeds (OSINT/commercial), APIs, and sharing communities.

2. Event Creation:

Each case/incident is stored as an Event.

Example: A phishing campaign → all related emails, IPs, domains are inside that event.

3. Attributes (IOCs):

Inside events, analysts add attributes (evidences) like IPs, domains, hashes, emails, files. And tags & taxonomies are applied (e.g., TLP for sensitivity, MITRE ATT&CK mapping).

4. Validation & Enrichment:

Using:

Warninglists → flag common false positives.

Correlation Engine → finds links between events (same IOC in multiple attacks).

Galaxies/Clusters → add context (e.g., threat actor, malware family).

5. Storage

All events & attributes are stored in the MISP database, available for search & filtering.

6. Sharing & Distribution:

Events can be shared with internal teams, trusted partners, or the wider community.

TLP/PAP rules control how widely information is shared.

7. Export & Integration:

Threat data can be exported (STIX, TAXII, CSV, JSON).

Integrated with IDS/IPS (Suricata, Snort), SIEM (Splunk, Wazuh), firewalls/EDRs.

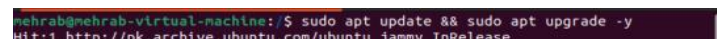
8. Collaboration (Feedback):

Other members can add comments, corrections, or new attributes. Data gets enriched over time, improving accuracy and value.

Environment Setup

1. System update:

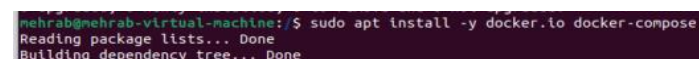
```
sudo apt update && sudo apt upgrade -y
```



```
mehrab@mehrab-virtual-machine:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://pk.archive.ubuntu.com/ubuntu jammy InRelease
```

2. Docker aur docker-compose install:

```
sudo apt install -y docker.io docker-compose
```



```
mehrab@mehrab-virtual-machine:~$ sudo apt install -y docker.io docker-compose
Reading package lists... Done
Building dependency tree... Done
```

3. MISP Docker repo clone:

git clone <https://github.com/harvard-itsecurity/misp-docker.git>

4. Open misp-docker container:

```
cd misp-docker
```

```
mehrab@mehrab-virtual-machine:~$ git clone https://github.com/MISP/misp-docker
cd misp-docker
Cloning into 'misp-docker'...
```

5. Permissions fix (important for volumes/logs):

```
sudo chown -R $USER:$USER ~/misp-docker
```

```
sudo chmod -R 775 ~/misp-docker
```

```
mehrab@mehrab-virtual-machine:~$ cd -
git clone https://github.com/MISP/misp-docker.git
cd misp-docker
Cloning into 'misp-docker'...
remote: Enumerating objects: 2541, done.
remote: Counting objects: 100% (750/750), done.
remote: Compressing objects: 100% (270/270), done.
remote: Total 2541 (delta 647), reused 506 (delta 480), pack-reused 1791 (from
)
Receiving objects: 100% (2541/2541), 571.65 KiB | 545.00 KiB/s, done.
Resolving deltas: 100% (1377/1377), done.
mehrab@mehrab-virtual-machine:~/misp-docker$ ls -l
total 108
drwxrwxr-x 3 mehrib mehrib 4096 13:27 29  docker-bake.hcl
-rw-rw-r-- 1 mehrib mehrib 4114 13:27 29  docker-compose.yml
-rw-rw-r-- 1 mehrib mehrib 12820 13:27 29  docker-compose.yml
```

6. cp template.env .env:

```
mehrab@mehrab-virtual-machine:~/misp-docker$ cp template.env .env
nano .env
```

Put URL for MISP interface

```
ADMIN_KEY=
# defaults to MISP's default (admin)
ADMIN_PASSWORD=
# defaults to 'passphrase'
GPG_PASSPHRASE=
# defaults to 1 (the admin user)
CRON_USER_ID=
# defaults to 'https://localhost'
# note: if you are exposing MISP on a non-standard port (i.e., the port is part
BASE_URL=http://localhost:80
# store settings in db except those that must stay in config.php. true/false, c
ENABLE_DB_SETTINGS=
# encryption key. defaults to empty string
```

Save Chanings.

7. Start container

```
docker compose pull
```

```
mehrab@mehrab-virtual-machine:~/misp-docker$ docker compose pull
```

```
9. Pulling 1/5
✓ redis Pulled                                4.3s
✓ misp-core Pulled                             3.8s
✓ misp-modules Pulled                         3.8s
✓ mail Pulled                                 4.4s
✓ db Pulled                                   4.3s
```

```
docker compose up -d
```

```
mehrab@mehrab-virtual-machine:~/misp-docker$ docker compose up -d
```

✓ Network misp-docker_default	Created	0.1s
✓ Container misp-docker-redis-1	Healthy	0.0s
✓ Container misp-docker-misp-modules-1	Healthy	0.0s
✓ Container misp-docker-mail-1	Started	1.2s
✓ Container misp-docker-db-1	Healthy	0.0s
✓ Container misp-docker-misp-core-1	Started	0.1s

8. Check container status

`docker ps -a`

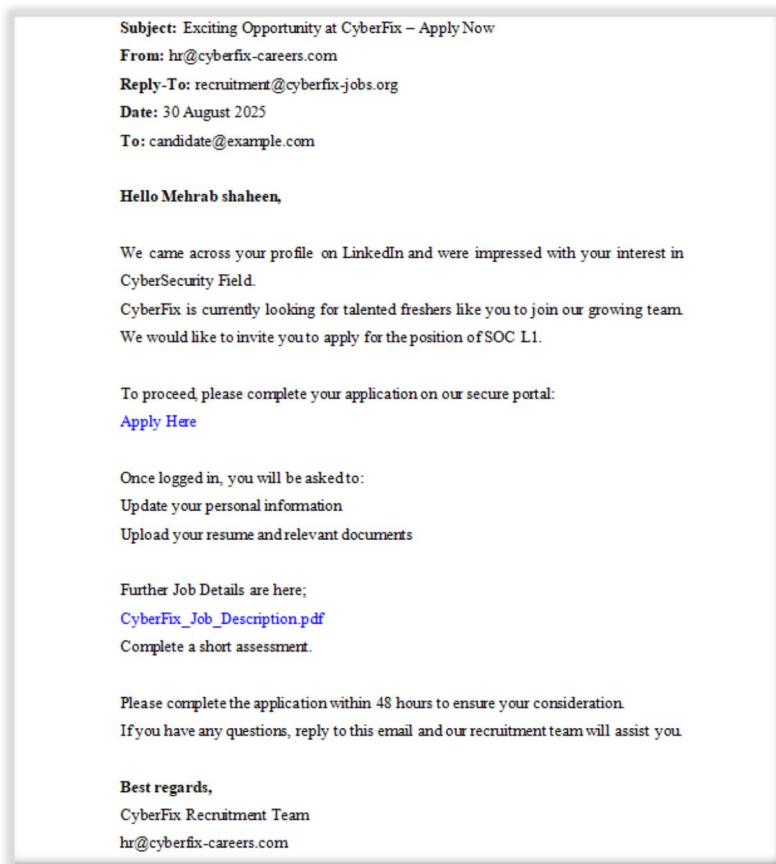
NAME	IMAGE	SERVICE	CREATED	STATUS	CONTR
misp-docker-db-1	marlabs:10.11	db	About a minute ago	Up About a minute (healthy)	"doc"
misp-docker-mail-1	txdotai/sntp	mail	About a minute ago	Up About a minute	"/bl"
misp-docker-misp-core-1	ghcr.io/misp/misp-docker/misp-core:latest	misp-core	About a minute ago	Up 58 seconds (healthy)	"/en"
misp-docker-misp-modules-1	ghcr.io/misp/misp-docker/misp-modules:latest	misp-modules	About a minute ago	Up About a minute (healthy)	"/us"
misp-docker-redis-1	valkey/valkey:7.2	redis	About a minute ago	Up About a minute (healthy)	"doc"

Use url in browser to access MISP.

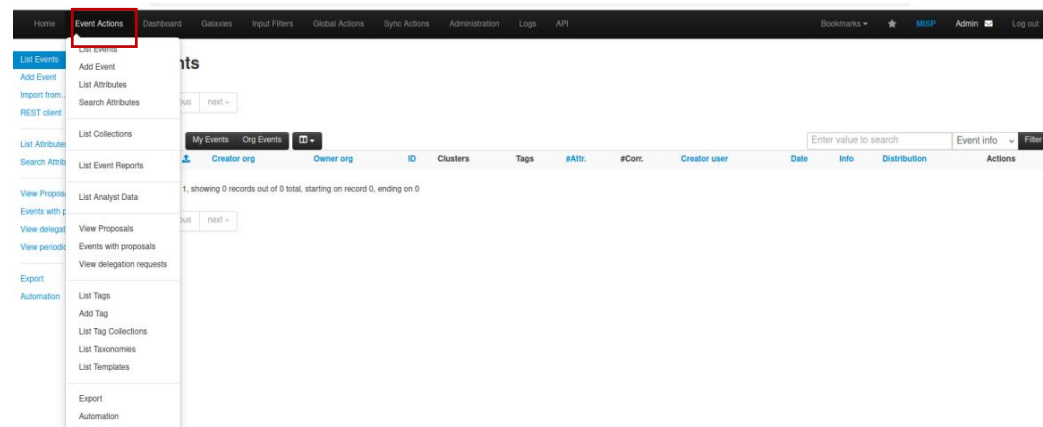
Case Study for understanding working of MISP

“Attackers are targeting LinkedIn users by abusing CyberFix Company’s name. They send fake job offer emails to steal credentials and deliver a malicious PDF attachment to gain access on user’s devices”. This scenario is used to demonstrate how MISP can capture, analyze, and share such threats.

Phishing Email sample:



Event Actions:



1. Event

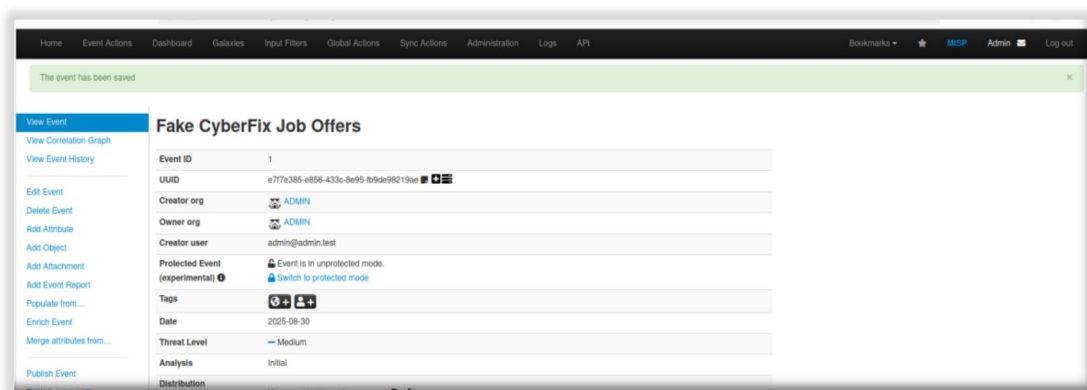
An “event” in MISP is like a case file that contains all the details about a specific cyber incident (e.g., phishing attack, malware campaign, ransomware, fraud attempt, etc.). It stores all collected evidences related cyber incident (IOC).

Note: I created event for a phishing email sample that I mentioned above.

A screenshot of the 'Add Event' form in the MISP web interface. The form is titled 'Add Event' and includes a notification at the top: 'The event created will be visible to the organisations having an account on this platform, but not synchronised to other MISP instances until it is published.' The form fields include: 'Date' (set to 2025-06-30), 'Distribution' (set to 'Your organisation only'), 'Threat Level' (set to 'Medium'), and 'Analysis' (set to 'Initial'). There is an 'Event Info' section with a text field containing 'Fake CyberFix Job Offers'. Below this is an 'Extends Event' section with a text field for 'Event UUID or ID. Leave blank if not applicable.' and a 'Submit' button. A sidebar on the left shows the 'Add Event' option selected under the 'Event Actions' menu.

Distribution Levels in MISP:

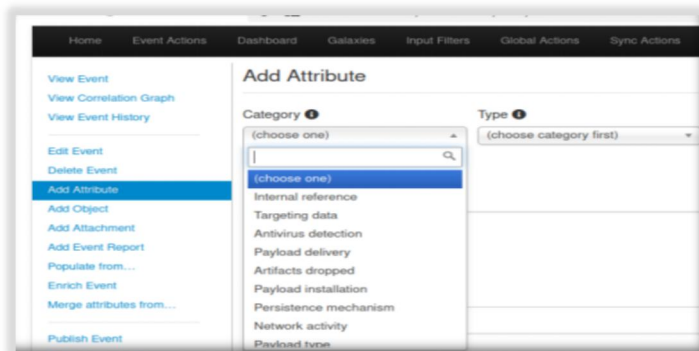
- **Your Organization Only:** Use when the data is sensitive and should remain internal.
- **This Community Only:** Use when sharing with a trusted group (like SOC team or specific partner).
- **All Communities:** Use when the data is safe to share globally, maximizing threat intelligence benefit.



2. Attributes

The “**Add Attribute**” option in MISP is used to enrich an event by adding technical indicators and details (IOCs). Attributes represent the individual pieces of data that describe a cyber incident. It help us:

- To document IOCs such as IPs, domains, hashes, or emails.
- To classify data into correct categories (e.g., Network, Payload, Email, External Analysis).
- To help detection tools (IDS/IPS, SIEM, EDR) use these indicators for monitoring and blocking.
- To share standardized threat intelligence across organizations.



Category: Groups the attribute by its role in the attack.

- **Network Activity:** Use for IP addresses, domains, URLs used in the attack.
- **Payload Delivery:** Use for file names, file hashes, malware samples.
- **Payload Installation:** Records how the malicious file or script was delivered/installed on the target.
- **Antivirus Detection:** Captures AV signatures and detection names for the malware sample.

- **Artifacts Dropped:** Documents files, registry keys, or traces created by malware after execution.

Attribute Types:

It shows what kind of data the indicator is. Each attribute has a type to describe the exact data. Examples:

- ip-src → source IP of the attacker
- ip-dst → target IP
- domain → malicious domain
- email-src → sender email address in phishing
- filename → malicious file name
- md5/sha1/sha256 → file hash of malware
- url → phishing or malicious site

2.1. Attribute - filename:

The screenshot shows the 'Add Attribute' form in a security dashboard. The form is titled 'Add Attribute' and has a left sidebar with navigation links. The main form area contains fields for Category (Payload delivery), Type (Filename), Distribution (Your organisation only), and Value (CyberFix_Job_Description.pdf). There is also a Contextual Comment field (attachment in phishing email) and checkboxes for Batch import, For Intrusion Detection System, and Disable Correlation. At the bottom, there are fields for First seen date, Last seen date, First seen time, and Last seen time, all with expected format HH:MM:SS.ssssss+TT:TT.

Practical View:

- I used the **Payload delivery** category with the type **Filename** because the attached file (**CyberFix_Job_Description.pdf**) represents the malicious content delivered to the victim.
- Recording this, will help in identifying and blocking suspicious file names in phishing campaigns.

2.2. Attribute - email-src:

The screenshot shows the 'Add Attribute' form in a web application. The 'Category' is set to 'Payload delivery' and the 'Type' is 'email-src'. The 'Distribution' is 'Your organisation only'. The 'Value' field contains 'hr@cyberfix-careers.com'. The 'Contextual Comment' field is empty. There are checkboxes for 'Batch import', 'For Intrusion Detection System', and 'Disable Correlation'. The 'First seen date' and 'Last seen date' are both set to '2025-10-30'. The 'First seen time' is '7:15:55 Am' and the 'Last seen time' is 'HH:MM:SS.ssssss+TT:TT'. The 'Expected format' for both times is 'HH:MM:SS.ssssss+TT:TT'. A 'Submit' button is at the bottom.

Practical View:

- I selected the “**Network activity**” category with the type “**Email-sr**”c since the source email address (**hr@cyberfix-careers.com**) is the primary indicator of the phishing attempt.
- It allows tracing the origin of the attack and spotting impersonation.

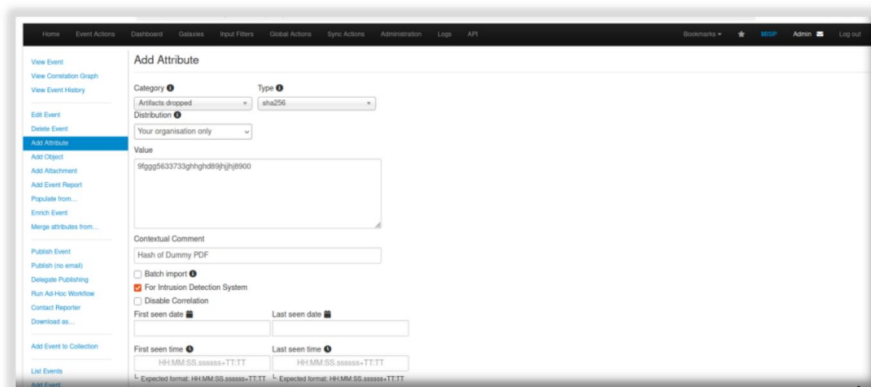
2.3. Attribute - email-reply-to:

The screenshot shows the 'Add Attribute' form in a web application. The 'Category' is set to 'Payload delivery' and the 'Type' is 'email-reply-to'. The 'Distribution' is 'Your organisation only'. The 'Value' field contains 'recruitment@cyberfix-jobs.org'. The 'Contextual Comment' field is empty. There are checkboxes for 'Batch import', 'For Intrusion Detection System', and 'Disable Correlation'. The 'First seen date' and 'Last seen date' are both set to '2025-08-30'. The 'First seen time' and 'Last seen time' are both set to 'HH:MM:SS.ssssss+TT:TT'. The 'Expected format' for both times is 'HH:MM:SS.ssssss+TT:TT'. A 'Submit' button is at the bottom.

Practical View:

- I used the **Network activity** category with the type “**Email-reply-to**” because attackers often use a different reply address (**recruitment@cyberfix-jobs.org**) to trick victims into communication.
- Documenting this helps detect mismatches between sender and reply channels.

2.4 Attribute - sha256:



Practical View:

- I chose the “**Artifacts dropped**”category with the type “**Hash**”as the file hash uniquely identifies the malicious attachment.
- This is critical for security teams to verify and detect the same malware sample across systems.

Note: in reality we have to verify hash of any file to confirm if its malicious.

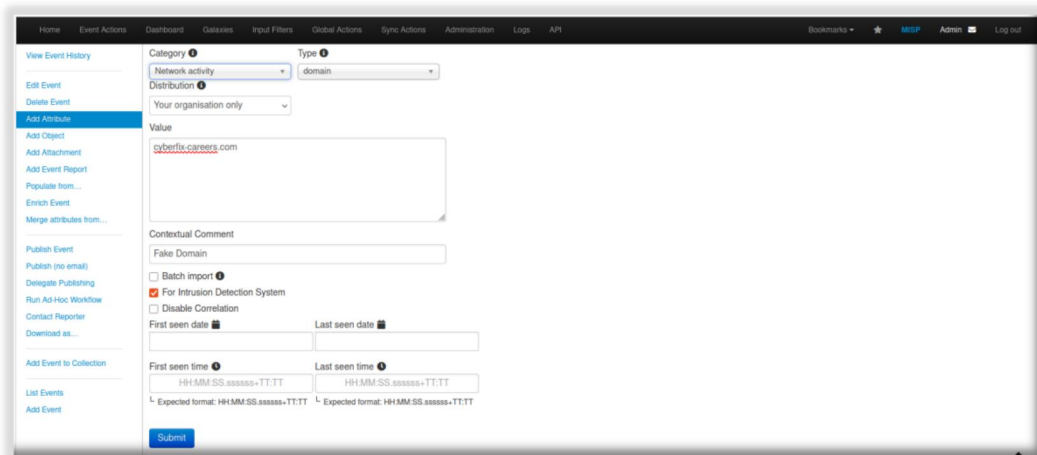
We use hash verification process:

1. Extract the hash of the suspicious file (e.g., from email attachment).
2. Submit the hash to public threat intelligence sources such as:
 - VirusTotal
 - MalwareBazaar (Abuse.ch)
 - ThreatFox

Key Points:

- If multiple security vendors flag the file as malicious, the hash is verified as a true malware sample.
- If the hash is not detected or flagged as clean, it does not belong to a malware family and should not be treated as malware.

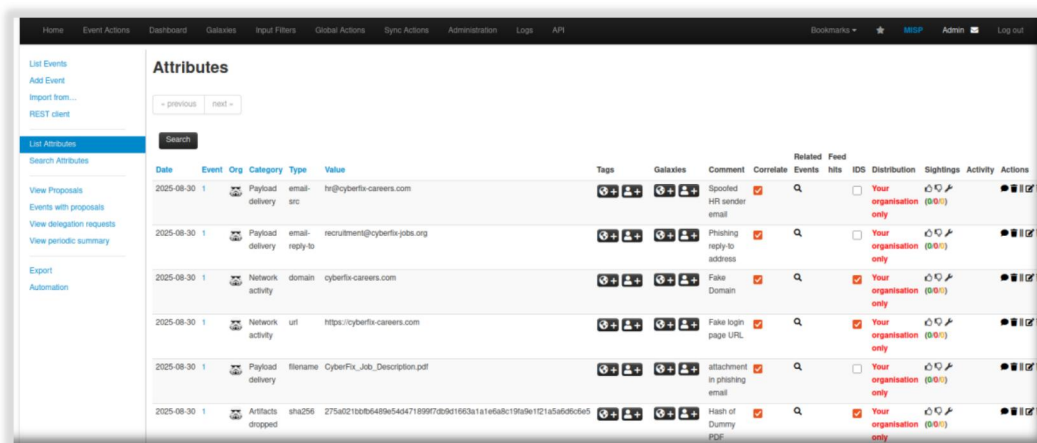
2.5. Attribute - domain:



Practical View:

- I used the “**Network activity**”category with the type “**Domain**” because phishing emails often use fake or look-alike domains (e.g., **cyberfix-careers.com**).
- Documenting the domain helps in detecting malicious infrastructure, blocking it at the network level, and preventing further phishing attempts.

List of attributes related Event ID 1:



Date	Event	Org	Category	Type	Value	Tags	Galaxies	Comment	Correlate	Events	Hits	IDB	Distribution	Sightings	Activity	Actions
2025-08-30	1		Payload delivery	email-src	hr@cyberfix-careers.com			Spooled HR sender email					Your organisation only (0/0)			
2025-08-30	1		Payload delivery	email-reply-to	recruitment@cyberfix-jobs.org			Phishing reply-to address					Your organisation only (0/0)			
2025-08-30	1		Network activity	domain	cyberfix-careers.com			Fake Domain					Your organisation only (0/0)			
2025-08-30	1		Network activity	url	https://cyberfix-careers.com			Fake login page URL					Your organisation only (0/0)			
2025-08-30	1		Payload delivery	filename	CyberFix_job_Description.pdf			attachment in phishing email					Your organisation only (0/0)			
2025-08-30	1		Artifacts dropped	sha256	27540210b0b6489e5404718997dbd1863a1a1e0a8c19a9e1121a5a6b0c0e5			Hash of Dummy PDF					Your organisation only (0/0)			

All evidence related attack are listed as attributes.

3. Add Analyst Opinion

Analyst can give opinion to provide expert commentary on the threat, explain the context, severity, and likely purpose of the attack.

Add new Opinion

Object Type: Opinion | Object UUID: [text box]

Distribution: Your organisation only

Authors: admin@admin.test

Opinion: 50 (slider) | Neutral 50

Comment: Verified phishing attempt targeting LinkedIn users. Payload delivery via email with malicious PDF.

Submit

Opinions Index

	Note	Opinion	Relationship	Parent Object Type	Target Object	Creator org	Created	Modified	Distribution	Comment	Opinion	Actions
1				Opinion		d51affbb-6684-4a45-9083-73e123dc531	2025-08-30 10:36:31	2025-08-30 10:36:31	Your organisation only	Verified phishing attempt targeting LinkedIn users. Payload delivery via email with malicious PDF.	Neutral 50	

Page 1 of 1, showing 1 records out of 1 total, starting on record 1, ending on 1

It helps other analysts to understand the reasoning behind the event and its attributes.

4. Add Attachment

The “**Add Attachment**” option is used to include malicious files or documents related to an event, such as PDFs, Word files, or executables.

Add Attachment(s)

Category: Payload delivery

Distribution: Your organisation only

Contextual Comment: malicious attachment PDF in phishing email

Browse... No files selected.

☒ Is a malware sample (encrypt and hash)

☐ Advanced extraction

Upload

Practical View:

- In the phishing email scenario, the **CyberFix_Job_Description.pdf** is added as an attachment to document the payload delivery mechanism.
- Recording attachments helps analysts analyze the file, extract indicators (hashes, macros), and share it with trusted partners for detection and mitigation.

5. Add Tags in Event

The **Custom Tag** option allows analysts to label an event or attribute with a user-defined, simple name that is not part of the standard taxonomies.

Custom tags help in quick filtering, internal tracking, and correlating similar events, making it easier for the team to manage and analyze threat intelligence within the **MISP** instance.

The screenshot shows the 'Add Tag' form in the MISP interface. The form includes fields for 'Name' (filled with 'fake-job-offer'), 'Colour' (filled with '#ff0000'), 'Restrict tagging to org' (set to 'ADMIN'), and 'Restrict tagging to user' (set to 'admin@admin.test'). There are checkboxes for 'Exportable', 'Hide Tag', and 'Enforce this tag to be used as local only'. A 'Submit' button is at the bottom.

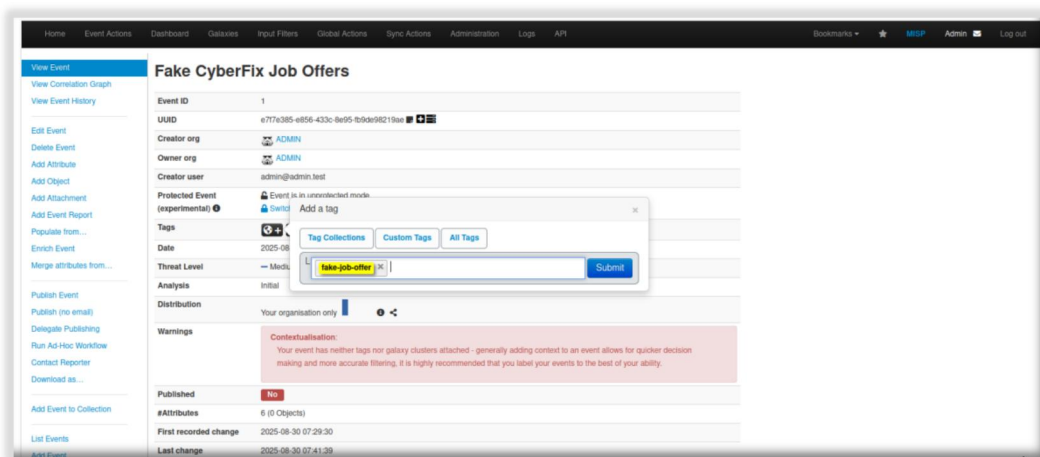
Practical View:

- I created a custom tag like **fake-job-offer** with customize color to clearly identify and categorize this specific campaign.

The screenshot shows the 'Tags' table in the MISP interface. The table has columns for ID, Exportable, Hidden, Local Only, Name, Restricted to org, Restricted to user, Taxonomy, Tagged events, Tagged attributes, Activity, Favourite, and Actions. The 'fake-job-offer' tag is highlighted in yellow.

ID	Exportable	Hidden	Local Only	Name	Restricted to org	Restricted to user	Taxonomy	Tagged events	Tagged attributes	Activity	Favourite	Actions
3	✓	✗	✗	estimative-language-likelihood-probability-almost-certain	✗	✗		0	0			
1	✓	✗	✗	estimative-language-likelihood-probability-likely	✗	✗		0	0			
4	✓	✗	✗	estimative-language-likelihood-probability-roughly-even-chance	✗	✗		0	0			
5	✓	✗	✗	estimative-language-likelihood-probability-unlikely	✗	✗		0	0			
2	✓	✗	✗	estimative-language-likelihood-probability-very-likely	✗	✗		0	0			
6	✗	✗	✓	fake-job-offer	✓	✓		0	0			

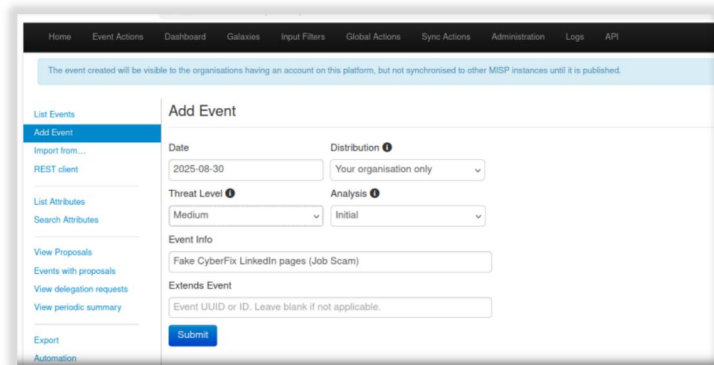
Use **View Event** option to use tag in your event.



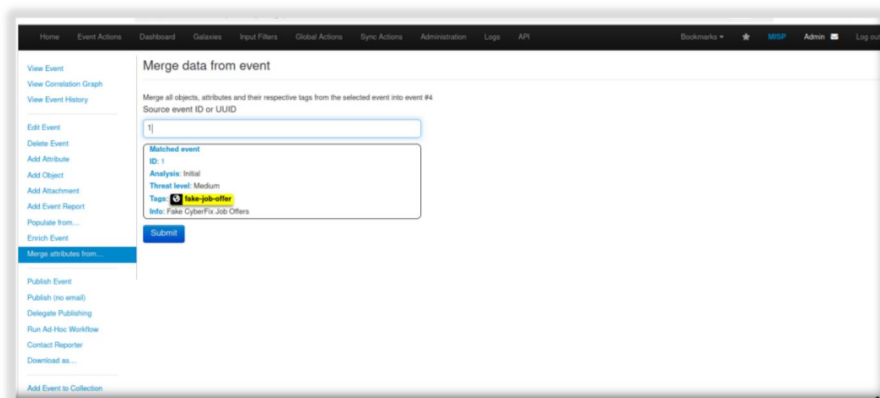
6. Merge Attributes for Event

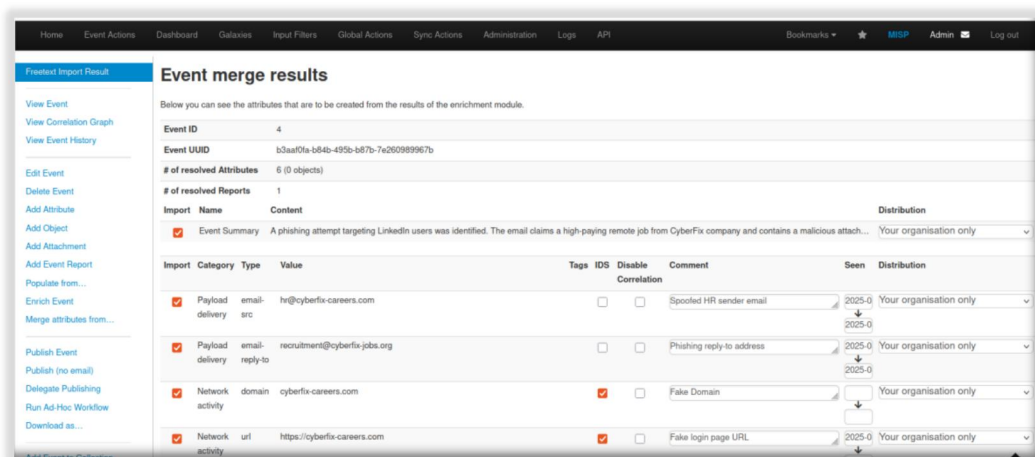
This option allows analysts to combine indicators (attributes) from one event into another related event. This helps in reducing duplication, improving correlation, and providing a complete view of the attack campaign across multiple events.

Create New Event:



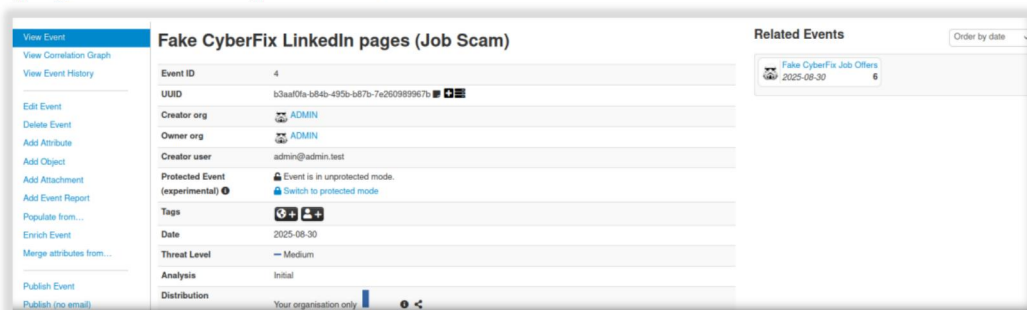
If a previous event contains similar attacker domains, sender emails, or malicious attachments, merging those attributes ensures all relevant IOCs are consolidated.





This new event is similar to Event ID 1.

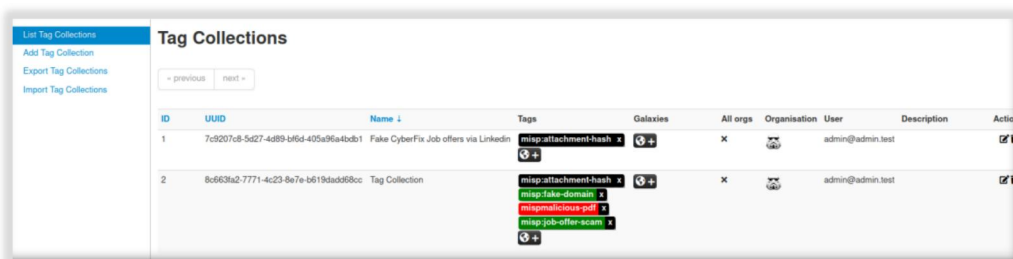
I merged all attributes of event 1 into it for providing a complete view of the attack campaign across multiple events.



7. Add Tag collection

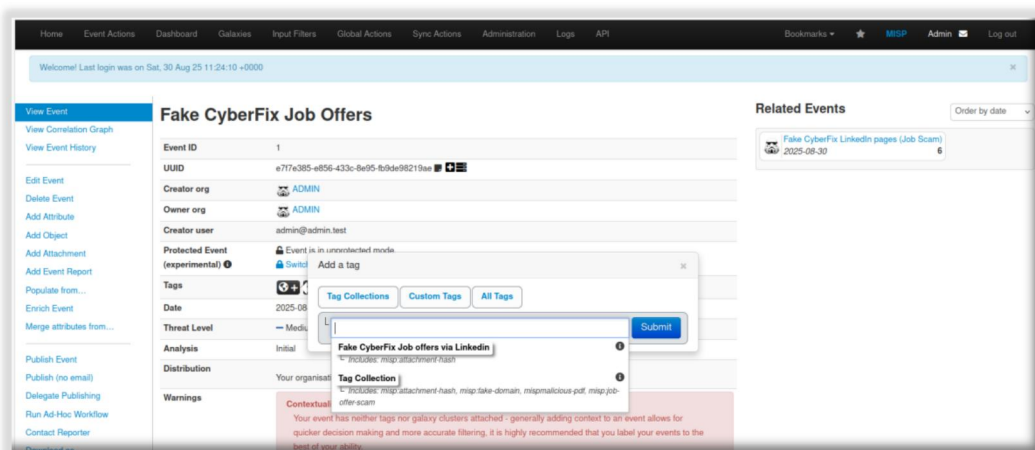
The “Tag Collection” feature allows analysts to apply multiple related tags to an event or attributes at once. Using a tag collection improves searchability, filtering, and understanding of complex events, helping teams quickly identify patterns or related incidents.

Note: First, create tags then just add them into tag collection.



I created tag collection to add multiple related tags in Event.

Use View Event to use Tag Collection:



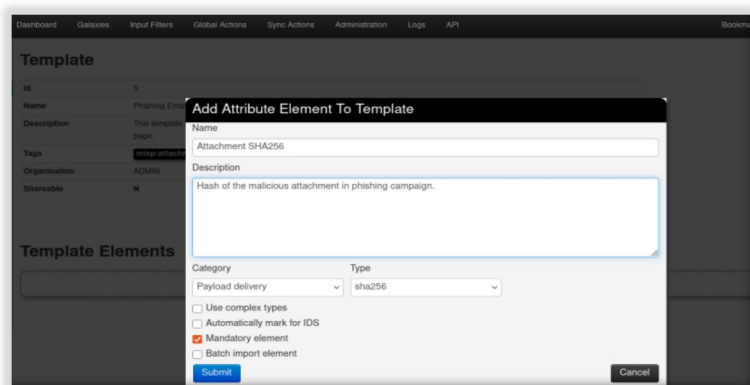
Event ID 1 with multiple custom tags:

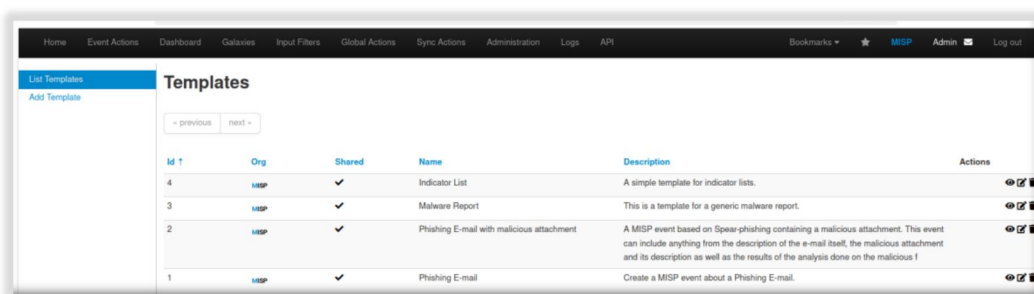
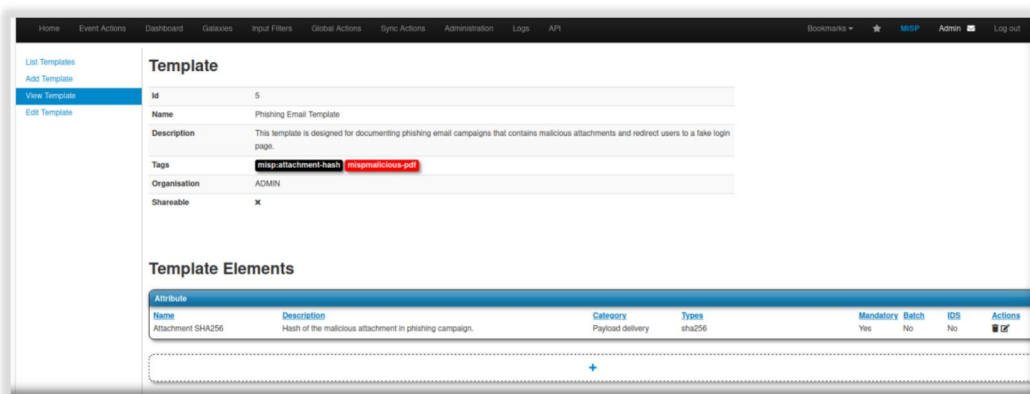


8. Build Intelligence Template

The “Add Template” option allows analysts to predefine a set of attributes, categories, and types that can be reused for similar events.

Using templates saves time, ensures consistency, and standardizes the way similar incidents are documented and shared across the team.



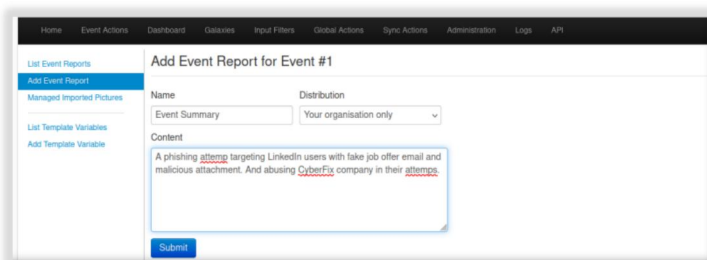


Practical View:

- I created phishing email template in MISP to standardize documentation of similar attacks. The template includes attributes **SHA256 hash** with related tags.
- Using this template ensures consistent recording of indicators, allows quick creation of new phishing events that may contain any malicious PDF or SHA256 Hash, and helps analysts identify, correlate, and share threat intelligence efficiently across related campaigns.

9. Add Report for Event:

The “**Add Report**” option allows analysts to attach a structured report or document to an event summarizing the incident, investigation details, and analysis. elligence with trusted teams or communities.

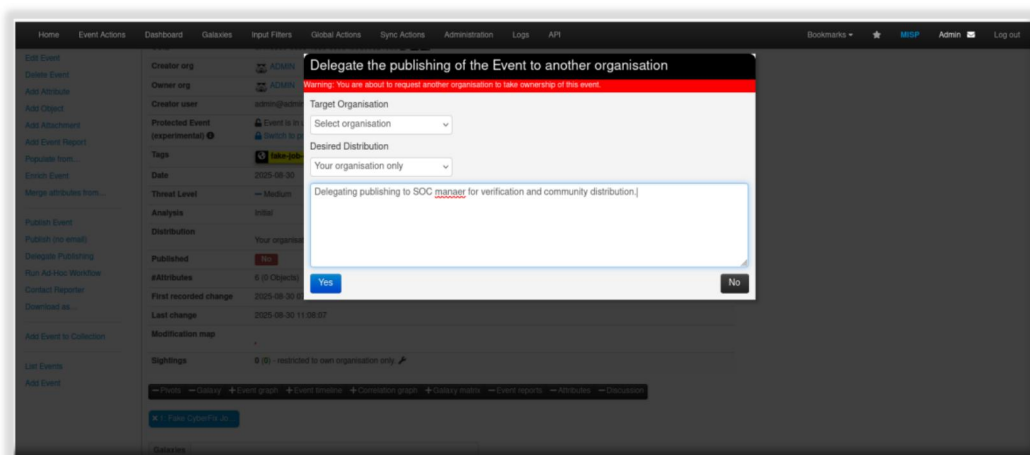


Practical View:

- In the phishing email case , a report can include the sender/recipient emails, malicious attachment, SHA256 hash, phishing URLs, and analyst notes.
- Adding a report provides context, documents findings for future reference, and facilitates sharing intelligence with trusted teams or communities.

10. Delegate the Publishing of the Event:

This option allows an analyst to assign another user or team to publish the event after it has been created.

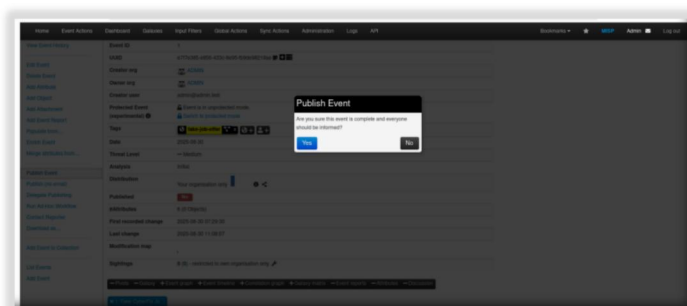


Practical View:

- In the phishing email scenario, delegation ensures that the event containing sensitive indicators (like attacker domains, malicious attachments, and SHA256 hashes) is reviewed and published by an authorized person, maintaining accuracy, proper sharing, and adherence to distribution policies.

11. Publish Event:

The “**Publish Event**” option makes an event available to other users or communities based on the selected distribution level.

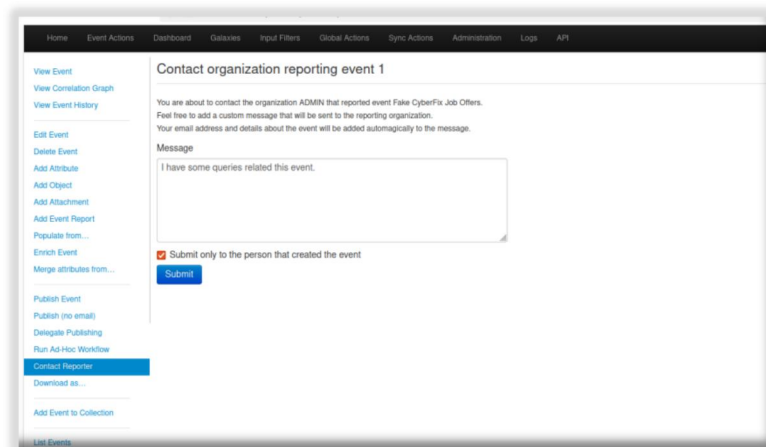


Practical View:

- In the phishing email scenario, publishing allows trusted teams or connected communities to receive and act on the indicators (malicious email, domains, attachment hash) for detection, blocking, or further investigation.
- It ensures that threat intelligence is shared securely and efficiently.

12. Contact Reporter:

The “**Contact Reporter**” option allows an analyst to reach out to the person or organization who submitted the event for clarification, additional details, or verification.

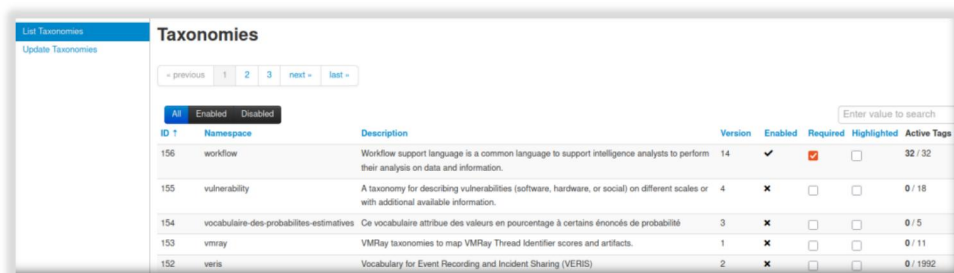


Practical View:

- Contacting the reporter help to confirm attributes like the malicious attachment, sender/reply-to emails, and phishing URLs, ensuring the event is accurate, complete, and reliable before sharing or taking action.

13. Add Taxonomies

Provides standardized tagging frameworks to classify and describe events or attributes. It adds context and meaning to indicators for easier analysis and sharing.



ID	Namespace	Description	Version	Enabled	Required	Highlighted	Active Tags
156	workflow	Workflow support language is a common language to support intelligence analysts to perform their analysis on data and information.	14	✓	☑	☐	32 / 32
155	vulnerability	A taxonomy for describing vulnerabilities (software, hardware, or social) on different scales or with additional available information.	4	✗	☐	☐	0 / 18
154	vocabulaire-des-probabilites-estimates	Ce vocabulaire attribue des valeurs en pourcentage à certains énoncés de probabilité	3	✗	☐	☐	0 / 5
153	vmray	VMRay taxonomies to map VmRay Thread Identifier scores and artifacts.	1	✗	☐	☐	0 / 11
152	veris	Vocabulary for Event Recording and Incident Sharing (VERIS)	2	✗	☐	☐	0 / 1992

- | Home | Event Actions | Dashboard | Calendar | Event Plans | Global Actions | Spgo Actions | Administration | Logs | API | Bookmarks | WSOP | Admin | Log out |
|------|---------------|-----------|----------|-------------|----------------|--|----------------|------|-----|-----------|------|-------|---------|
| 12 | ✓ | ✗ | ✗ | ✗ | 0 | unknown_classification未知category_Unknown | | ✗ | ✗ | 5 | 0 | 0 | ✗ |
| 11 | ✓ | ✗ | ✗ | ✗ | 0 | 未知gateway Street action_Unknown | | ✗ | ✗ | 1 | 0 | 0 | ✗ |
| 52 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 53 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 54 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 55 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 10 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 2 | 0 | 0 | ✗ |
| 51 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 56 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 8 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 40 | 0 | 0 | ✗ |
| 7 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知 | | ✗ | ✗ | 20 | 0 | 0 | ✗ |
| 45 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | 0 | 0 | 0 | ✗ |
| 46 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |
| 44 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |
| 47 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |
| 48 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |
| 49 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |
| 30 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |
| 39 | ✓ | ✗ | ✗ | ✗ | 0 | 未知未知未知未知 | | ✗ | ✗ | workflow | 0 | 0 | ✗ |

The “**Add Collection**” feature in MISP lets analysts group related attributes or events into a single set for easier management.

Home

Event Actions

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Galleries

Input Filters

Global Actions

Sync Actions

Administration

Logs

API

Bookmarks +

★

MSP

Admin

⌵

Log out

List Collections

Add Collection

Add new collection

Create collections to organise data shared by the community into buckets based on commonalities or as part of your research process. Collections are first class citizens and adhere to the same sharing rules as for example events do.

Name

LinkedIn Fake Job Offer Phishing Emails

Type

campaign

Description

This collection contains phishing emails, domains, URLs, and payload hashes related to the LinkedIn fake job offer campaign observed in August 2025

Distribution

Sharing Group

Your organisation only

SOC Analysts

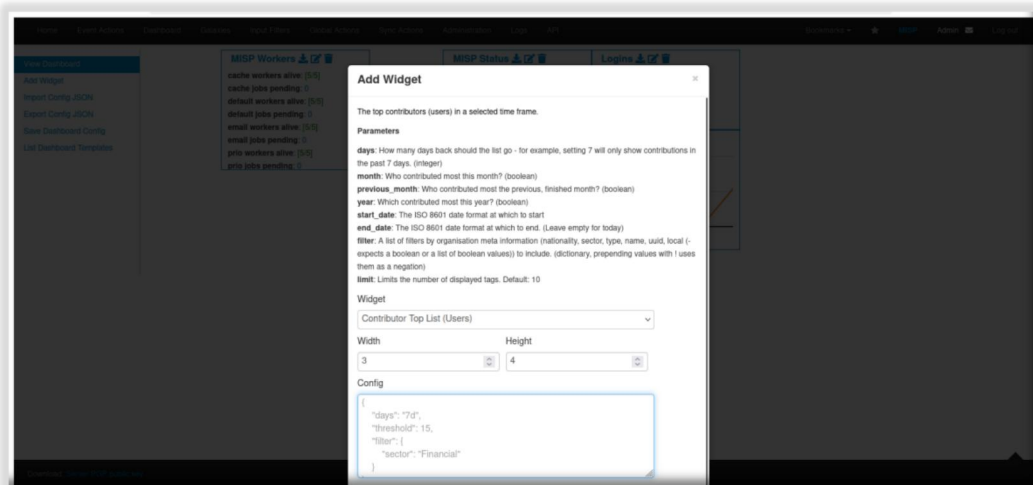
Submit

In the phishing email scenario, attributes like the sender email, reply-to address, malicious PDF attachment, SHA256 hash, and phishing domain are grouped into a collection.

- See all indicators related to a campaign in one place.
- Correlate attributes across events.
- Simplify analysis and reporting.
- Share organized threat intelligence with trusted teams or communities.

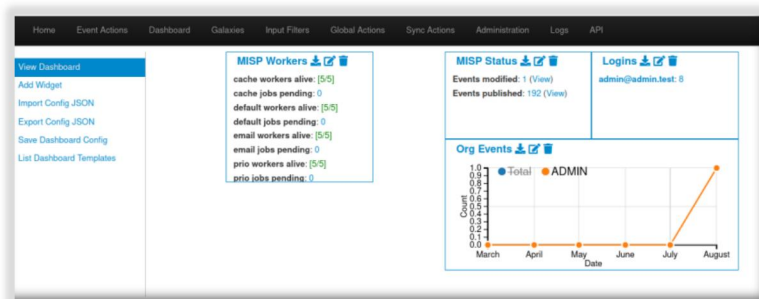
15. Threat Intelligence Dashboard

The Dashboard provides a centralized overview of all activities and data within MISP, helping analysts monitor, manage, and act on threat intelligence efficiently.



Usage:

1. **Attribute Tracking:** Displays recently added indicators (IPs, domains, emails, file hashes) and any updates to existing events.
2. **Statistics & Trends:** Provides insights into the volume of events, distribution levels, or types of threats being reported over time.
3. **Quick Access:** Shortcuts to add new events, attributes, tags, or perform searches across all events.



Practical View:

MISP Workers

- Shows the status of background processes (workers) handling tasks like attribute processing, event correlation, and notifications.
- Ensures all automated MISP tasks are running smoothly and highlights any failed or delayed jobs.

Logins

- Displays recent user login activity on the MISP platform.
- Helps track analyst access, monitor unusual logins, and maintain accountability for actions performed in MISP.

Organization Events (Org Events)

- Provides a summary of events created or managed by your organization.
- Helps analysts quickly see the volume, status, and distribution of internal events.

MISP Status

- Shows the overall health and operational status of the MISP instance.
- Ensures the platform is functioning properly, including database connectivity, workers, and API integrations.

16. Add Galaxy

A Galaxy in MISP is a predefined collection of related threat intelligence. It represents high-level entities such as:

- Threat actors (APT groups, cybercriminal groups)
- Malware families (e.g., Emotet, TrickBot)
- Tools or campaigns used by attackers

Makes reporting structured and consistent, so everyone can understand who or what is behind the attack.

Assigning a Galaxy to an event helps in classifying the attack type and giving context to other analysts or SOC teams.

Galaxy Index:

Serves as the master catalog of all Galaxies available in MISP. Analysts search the index to identify the correct threat actor, malware family, or campaign related to an observed attack.

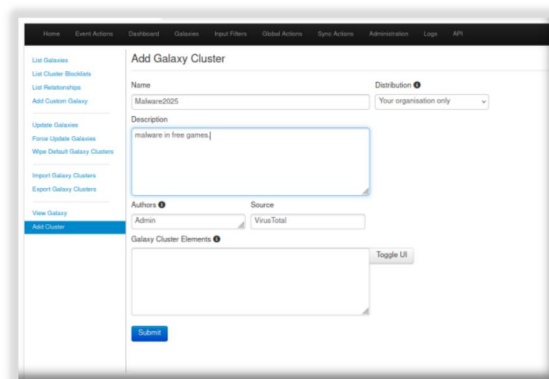
ID	Icon	Name	Version	Namespace	Description	Enabled	Local Only	Actions
99	📁	UKHSA Culture Collections	1	gov.uk	UK Health Security Agency Culture Collections represent deposits of cultures that consist of expertly preserved, authenticated cell lines and microbial strains of known provenance.	✓	✗	🔍 📄 🗑️
98	✈️	UAVs/UCAVs	2	miap	Unmanned Aerial Vehicles / Unmanned Combat Aerial Vehicles	✓	✗	🔍 📄 🗑️
97	🔧	Tool	3	miap	Threat actors tools is an enumeration of tools used by adversaries. The list includes malware but also common software regularly used by the adversaries.	✓	✗	🔍 📄 🗑️
96	📁	Threat Matrix for storage services	1	microsoft	Microsoft Defender for Cloud threat matrix for storage services contains attack tactics, techniques and mitigations relevant storage services delivered by cloud providers.	✓	✗	🔍 📄 🗑️
95	👤	Tidal Technique	1	tidal	Tidal Technique Galaxy	✓	✗	🔍 📄 🗑️
94	📁	Tidal Tactic	1	tidal	Tidal Tactic Galaxy	✓	✗	🔍 📄 🗑️
93	📁	Tidal Software	1	tidal	Tidal Software Galaxy	✓	✗	🔍 📄 🗑️
92	📁	Tidal References	1	tidal	Tidal References Galaxy	✓	✗	🔍 📄 🗑️
91	👤	Tidal Groups	1	tidal	Tidal Groups Galaxy	✓	✗	🔍 📄 🗑️
90	📁	Tidal Campaigns	1	tidal	Tidal Campaigns Galaxy	✓	✗	🔍 📄 🗑️

Practical View:

- I created a custom Galaxy named **“Fake Game Malware Campaign”** to attach the phishing email event, to indicate the attack could be linked to a known malware campaign.

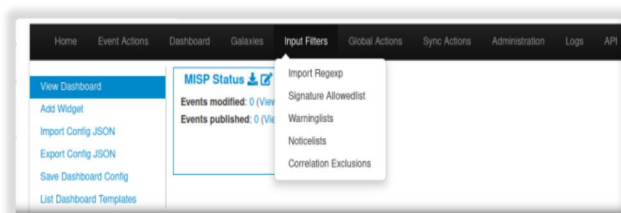
Property	Value
Galaxy ID	100
Name	Fake Game Malware Campaign
Namespace	custom
UUID	629a424-408-444a-887-47802770a30
Description	Campaign distributing malware via free games.
Version	179602179
Created	2025-08-31 08:22:56
Modified	2025-08-31 08:22:56
Enabled	True
Local Only	No
Distribution	Your organisation only
Owner Organisation	👤
Creator Organisation	👤

A Cluster represents a specific instance, version, or wave of the malware campaign.



I created a Cluster inside the Galaxy called **“Malware 2025”**.

17. Add Input Filters



Used to validate, sanitize, or block invalid data before ingestion. It maintains IOC quality and avoids storing junk/malformed indicators. Admin defines filters (blocklist, allowlist, regex validation). Incoming data from feeds/API must pass these rules.

17.1. Input filters - Regexp:

It uses Regular Expressions (Regex) to validate observables before they are stored. It ensures only properly formatted, clean data enters MISP. Prevents invalid or malformed observables (e.g., wrong email or incomplete IP). And maintains high data quality across threat intelligence.

Id	Regexp	Replacement	Type	Actions
1	./ProgramData./	%ALLUSERSPROFILE%	ALL	[icon] [icon]
2	./Documents and Settings.All Users./	%ALLUSERSPROFILE%	ALL	[icon] [icon]
3	./Program Files.Common Files./	%COMMONPROGRAMFILES%	ALL	[icon] [icon]
4	./Program Files (x86).Common Files./	%COMMONPROGRAMFILES(x86)%	ALL	[icon] [icon]
5	./Users/(.*)/AppData/Local/Temp./	%TEMP%	ALL	[icon] [icon]
6	./ProgramData./	%PROGRAMDATA%	ALL	[icon] [icon]
7	./Program Files./	%PROGRAMFILES%	ALL	[icon] [icon]
8	./Program Files (x86)./	%PROGRAMFILES(x86)%	ALL	[icon] [icon]
9	./Users.Public./	%PUBLIC%	ALL	[icon] [icon]
10	./Documents and Settings/(.*)/Local Settings/Temp./	%TEMP%	ALL	[icon] [icon]
11	./Users/(.*)/AppData/Local/Temp./	%TEMP%	ALL	[icon] [icon]
12	./Users/(.*)/AppData/Local./	%LOCALAPPDATA%	ALL	[icon] [icon]
13	./Users/(.*)/AppData/Roaming./	%APPDATA%	ALL	[icon] [icon]
14	./Users/(.*)/Application Data./	%APPDATA%	ALL	[icon] [icon]

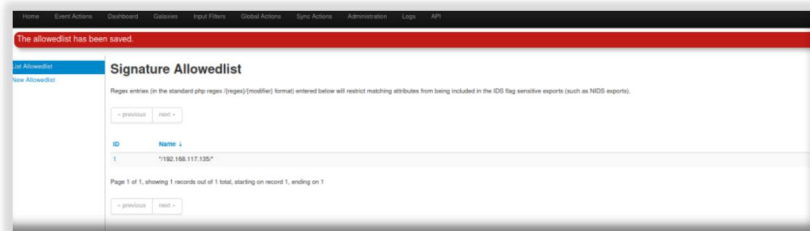
Admin defines regex rules under Administration → Input Filters.

When an analyst adds an observable (IP, domain, email, hash), MISP checks it against regex.

- If it matches regex → accepted into the database.
- If it fails regex → rejected with a warning/error message.

17.2. Signature Allowedlist:

A whitelist of trusted signatures/observables that should not trigger alerts. It plays a role in reducing false positives in SIEM/IDS/IPS exports.

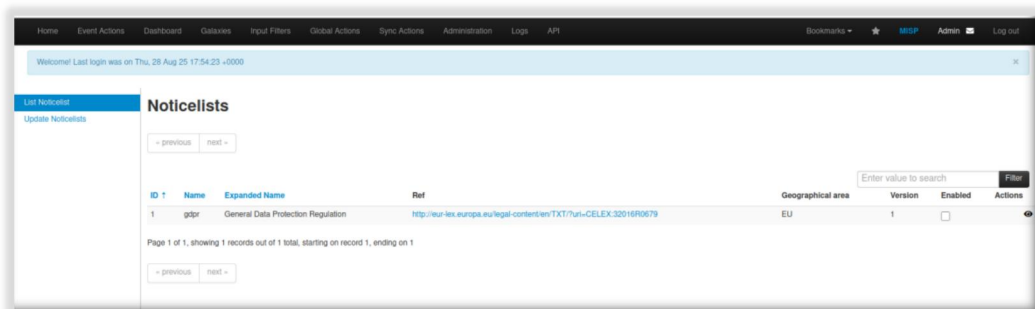


Admin maintains a list of safe values (like scanner IPs). During IOC export, these are excluded.

17.3. Noticelists:

Provide context-sensitive reminders to analysts when adding data. It ensures compliance with legal/regulatory rules (e.g., GDPR).

When certain attributes are added (like PII), a notice appears reminding about handling policies.



Practical View:

- When adding the victim's email address from the phishing case, Noticelists reminded analysts about GDPR restrictions.

17.4. Warninglists:

Built-in lists of known safe/false-positive values (e.g., Alexa top 1M domains, Google DNS). It avoids wasting time on noise; analysts can focus on real threats.

When adding an attribute, MISP checks against Warninglists. If matched, a warning flag appears.

Example:

The phishing email footer contained a microsoft.com link. Warninglists flagged it as benign, so analysts ignored it.

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Link to overview

Warninglists

Search in Warninglists

Warninglists

- previous

1

2

next - last -

All

Enabled

Disabled

Enter value to search

Filter

ID	Name	Version	Description	Category	Type	Entries	Details	Enabled	Actions
91	List of known Zscaler IP address ranges	20230710	Zscaler IP address ranges (https://nfdz.zscaler.com/api/zscaler/net/hub/cidr/json?reqid=1)	False positive	cidr	118		✓	x
92	List of known Wikimedia address ranges	20230710	Wikimedia address ranges (http://noc.wikimedia.org/internet/private.pptx.txt)	False positive	cidr	46		✓	x
93	List of known domains to know about in Open Whois my IP domains	20230710	Event contains one or more entries of known Whois my IP domains	False positive	hostname	212		✓	x
94	Specialized list of IPv4 addresses belonging to common VPN providers and datacenters	20230402	Specialized list of IPv4 addresses belonging to common VPN providers and datacenters	False positive	cidr	1260		✓	x
87	Specialized list of vpn-qvd addresses belonging to common VPN providers and datacenters	20230710	Specialized list of vpn-qvd addresses belonging to common VPN providers and datacenters	False positive	cidr	9999		✓	x
95	List of known URL Shorteners domains	21	Event contains one or more entries of known Shorteners domains	False positive	hostname	175		✓	x
96	University domains	20230710	List of University domains from https://raw.githubusercontent.com/HipCity/university-domains/master/univ_universities_and_domains.json	False positive	string	16477		✓	x
84	ciroo-ubuntu16-knowledge-gw0	20230710	Cisco Umbrella knowledge in IPv4	False positive	cidr	9		✓	x
83	ciroo-ubuntu16-knowledge-gw4	20230710	Cisco Umbrella knowledge in IPv4	False positive	cidr	6		✓	x
82	ciroo-ubuntu16-knowledge-hostname	20230710	Umbrella knowledge hostnames	False positive	hostname	5		✓	x
81	Top 10K most used sites from Tranco	20230115	Event contains one or more entries from the top 10K most used sites (https://transo.net/en/)	False positive	hostname	10000		✓	x
80	Top 1,000,000 most used sites from Tranco	20230115	Event contains one or more entries from the top 1,000,000 most used sites (https://transo.net/en/)	False positive	hostname	1000000		✓	x
79	TLDs as known by IANA	20230710	Event contains one or more TLDs as attribute with an IDS flag set	False positive	string	1440		✓	x
78	Hostnames that are often included in IOC lists but are false positives.	1	Hostnames that are often included in IOC lists but are false positives.	False positive	string	2644		✓	x
77	List of known Tornode Cloud Services IPv4s	20230710	Tornode IPv4 Cloud Sensor addresses used for scanning Internet facing infrastructure	False positive	cidr	17		✓	x
76	List of known Tornode Cloud Services IPv4s	20230710	Tornode IPv4 Cloud Sensor addresses used for scanning Internet facing infrastructure	False positive	cidr	39		✓	x
75	List of known Startpage CDN IP ranges	20230402	List of known Startpage (Highlander) CDN IP ranges (https://support.startpage.com/en/articles/36001191686-Whitelist-Startpage-IP-Ranges)	False positive	cidr	247		✓	x
74	List of known SMTP sending IP ranges	20230710	List of IP ranges for known SMTP servers.	False positive	cidr	1016		✓	x
73	List of known SMTP receiving IP ranges	20230710	List of IP addresses for known SMTP servers.	False positive	cidr	305		✓	x
	List of known archives	2	List of known archives	False positive	cidr	107		✓	x

pcalhost-8443/warninglists/index#

18. Add User

To onboard new analysts, researchers, or external threat intel partners into MISP. It makes the platform collaborative; multiple SOC/CTI analysts can contribute.

	Home	Event Actions	Databases	Collections	Ingot Filters	Globed Actors	Syntic Actors	Administration	Logs	APl
--	------	---------------	-----------	-------------	---------------	---------------	---------------	----------------	------	-----

Add User
Add Users
Pending Registrations
User settings
Site Settings
Contact Users
Add Organisation
List Organisations
Add Role
List Roles
Screen Settings & Maintenance
Update Progress
Jobs
Scheduled Tasks
Event Block Rules
Booklist Event
Manage Event Booklists
Booklists Organisation
Manage Org Booklists
Booklists Signings
Manage Signing Booklists

Add Admin User Form:

Email: admin@admin.test

[x] Set Password Confirm Password

Password: ****

Organisation: ADMIN

Role: NDIS SID

User: admin

SynC user for: Not bound to a server

PGP key:

```
+vsoDghwKqWzBdHggGngFjEE+YtOIZgP/TbxLLLSQANDB+QofA3JdqMC  
EawTgwZrXauSdr+QuV/rjpmL+MwNHzkUyI+jPgR+xIcABkd3mjdveNVe  
6TwAGnF/zljk/NorJoabJdB/gY/VbaalhwagZF/SYSR/vZrvPhaoSD  
gtlxTZwgZSp/WkfNF23oKSS3LMBeuBgWZ+NforULAwENldgaJDSGBBY  
HksXnmSR+zPywskwkdgguWcuNgpZf2867/YouLapd/yacWkbwh/BFAHQZGY  
Gtyh/fBy/ZfwH4enSL/Coclvgr220BA0BRT7MSgPYTCR3angvsnvbnkgDGLd
```

Fatch PGP Key [?]

[x] Receive email alerts when events are published

[x] Receive email alerts from "Contact reporter" requests

☐ Immediately disable this user account

[x] Send credentials automatically

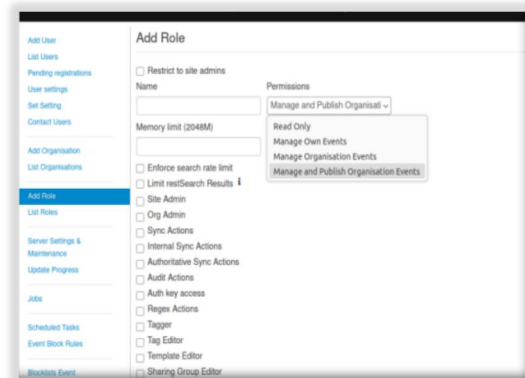
Create user.

Practical View:

- A new SOC analyst was added to investigate phishing email headers and enrich observables.

19. Add Role

To enforce role-based access control (RBAC). To prevent misuse, ensures least privilege access. For example, not everyone can delete events.



ID	Default	Name	Permission	Site Admin	Org Admin	Sync Actions	Internal Sync Actions	Authoritative Sync Actions	Audit Actions	Auth key access	Regex Actions	Tagger	Tag Editor	Template Editor	Sharing Group Editor	Memory Limit	Max execution time	Searches / 15 min
1	☐	admin	Manage and Publish Organisation Events	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	2048M	300 s	Unlimited
2	☐	Org Admin	Manage and Publish Organisation Events	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	2048M	300 s	Unlimited
3	☒	User	Manage Organisation Events	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	2048M	300 s	Unlimited
4	☐	Publisher	Manage and Publish Organisation Events	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	2048M	300 s	Unlimited
5	☐	Sync user	Manage and Publish Organisation Events	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	2048M	300 s	Unlimited
6	☐	Read Only	Read Only	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	2048M	300 s	Unlimited

Practical View:

- The analyst investigating the phishing case was given read-only access, so they could review but not modify evidence.

20. Check Audit Logs:

To track user actions across the system. It ensures accountability; investigators can trace back who added, edited, or deleted what.

Every user activity (login, data entry, exports) is recorded. Admin can view these logs for compliance.

Created	User	IP	Org	Action	Model	Title	Change
2025-08-30 11:08:07	admin@admin.test	172.18.0.1	US	Delete	EventReport #1	Event Summary	uuid: "380ed73a-dc1-4b99-8b93-500b623b7b11" → X event_id: 1 → X name: "Event Summary" → X content: "A phishing attempt targeting LinkedIn users with take job offer e..." → X
2025-08-30 11:07:25	admin@admin.test	172.18.0.1	US	Add	EventReport #2	Event Summary	name: → "Event Summary" content: → "A phishing attempt targeting LinkedIn users was identified. The..." event_id: → 1 uuid: → "1e566e49-3118-480d-8b93-01390b2f500f"
2025-08-30 11:00:30	admin@admin.test	172.18.0.1	US	Add	EventReport #1	Event Summary	name: → "Event Summary" content: → "A phishing attempt targeting LinkedIn users with take job offer e..." event_id: → 1 uuid: → "380ed73a-dc1-4b99-8b93-500b623b7b11"
2025-08-30 10:36:31	admin@admin.test	172.18.0.1	US	Add	Option #1	Option #1	object_type: → "Option" authors: → "admin@admin.test" option: → 36

Practical View:

- Audit logs showed which analyst uploaded the phishing PDF hash into the event.

21. Check Application Logs:

Used to keep track of all system-level and technical activities. It helps in troubleshooting, debugging, and ensuring feeds/syncs are working fine and to records errors, feed pulls, background jobs, sync activity..

Id	Created	IP	Email	Org	Model	Model ID	Action	Title	Change
1066	2025-08-30 11:35:17		admin@admin.test	SYSTEM	User	1	email	Email from admin@admin.test to admin@admin.test sent, titled "MISP New sign-in".	
1065	2025-08-30 11:24:10	172.18.0.1	admin@admin.test	SYSTEM	User	1	email	Email to admin@admin.test sent, titled "MISP New sign-in".	
1064	2025-08-30 11:24:10	172.18.0.1	admin@admin.test	ADMIN	User	1	login	User (1): admin@admin.test	["user_agent":"Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:142.0) Gecko/20100101 Firefox/142.0", "ip":"172.18.0.1", "geoip":"None", "ua_pattern":"mozilla/5.0 (linux) gecko", "platform":"Linux", "ua...]
1063	2025-08-30 10:15:40	172.18.0.1	admin@admin.test	SYSTEM	User	1	email	Email to admin@admin.test sent, titled "MISP New sign-in".	
1062	2025-08-30 10:15:40	172.18.0.1	admin@admin.test	ADMIN	User	1	login	User (1): admin@admin.test	["user_agent":"Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:142.0) Gecko/20100101 Firefox/142.0", "ip":"172.18.0.1", "geoip":"None", "ua_pattern":"mozilla/5.0 (linux) gecko", "platform":"Linux", "ua...]
1061	2025-08-30 08:22:37	172.18.0.1	admin@admin.test	SYSTEM	User	1	email	Email to admin@admin.test sent, titled "MISP New sign-in".	
1060	2025-08-30 08:22:37	172.18.0.1	admin@admin.test	ADMIN	User	1	login	User (1): admin@admin.test	["user_agent":"Mozilla/5.0 (X11; Ubuntu; Linux x86_64; rv:142.0) Gecko/20100101 Firefox/142.0", "ip":"172.18.0.1", "geoip":"None", "ua_pattern":"mozilla/5.0 (linux) gecko", "platform":"Linux", "ua...]
1059	2025-08-30 07:40:26	172.18.0.1	admin@admin.test	ADMIN	Attribute	0	add	Attribute dropped due to validation for Event 1 failed: Artifacts dropped sha256 9dgg5633733grhghd99ygh89006666ggfuff	Validation errors: ["value":"Checksum has an invalid length or format (expected: 64 hexadecimal character value or select type 'other')"]; Full Attribute: {"category":"Artifacts", "type":"sha256", "distribution":"O", "value":"9dgg5633733grhghd99ygh89006666ggfuff", "comment":"PDF: Security report 2025-08-30 11:35:17", "source":"https://www.misp-project.org/docs/en/latest/feeds/feeds.html#event-attributes"}.

Practical View:

- Logs confirmed that observables from phishing report feed are ingested successfully.

22. Add Sharing Group

Used to define who can access specific threat intelligence data (events, attributes, observables). It protects sensitive IOCs from being leaked to untrusted or irrelevant organizations.

Name	Is local	Can extend
ADMIN	✓	✓

Name	URL	All orgs
Local instance	https://localhost:8443	✗

Practical View:

- IOC of phishing campaign is only shared with the SOC Analysts of CyberFix.